

# Algebra Essentials



## Evaluating expressions

- 1 If  $x = 4$ , evaluate  $3x + 7$ .
  - 2 If  $a = 5$  and  $b = 2$ , evaluate  $a^2 - 3b$ .
  - 3 If  $x = -3$ , evaluate  $2x^2 + 5x$ .
  - 4 If  $p = 6$  and  $q = -2$ , evaluate  $\frac{p - q}{q + 6}$ .
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## Solving linear equations

- 5 Solve  $2x + 5 = 17$ .
  - 6 Solve  $5x - 8 = 27$ .
  - 7 Solve  $3(x - 4) = 21$ .
  - 8 Solve  $\frac{x}{4} + 3 = 10$ .
  - 9 Solve  $7x + 3 = 4x + 18$ .
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## Expanding and simplifying

- 10 Expand  $(x + 3)(x + 5)$ .
  - 11 Expand  $(x - 4)(x + 6)$ .
  - 12 Expand  $(2x + 1)(x - 3)$ .
  - 13 Simplify  $3(2x - 5) - 2(x - 4)$ .
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## Factoring

- 14 Factor  $x^2 + 9x + 20$ .
- 15 Factor  $x^2 - 2x - 15$ .

16 Factor  $x^2 - 49$ .

17 Factor  $6x + 15$ .

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### Solving quadratic equations by factoring

18 Solve  $x^2 + 5x + 6 = 0$ .

19 Solve  $x^2 - 7x + 12 = 0$ .

20 Solve  $x^2 - 16 = 0$ .

21 Solve  $x^2 + 2x - 15 = 0$ .

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### Systems of two equations

22 Solve the system:  $x + y = 10$ ,  $x - y = 4$ .

23 Solve the system:  $2x + y = 11$ ,  $x - y = 1$ .

24 Solve the system:  $x = 3y - 1$ ,  $2x + y = 19$ .

25 Solve the system:  $4x - y = 10$ ,  $2x + y = 8$ .

26 Solve the system:  $5x + 3y = 21$ ,  $2x - 3y = 0$ .

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### Laws of exponents

27 Simplify  $a^5 \times a^3$ .

28 Simplify  $\frac{x^9}{x^4}$ .

29 Simplify  $(y^3)^4$ .

30 Simplify  $2x^3 \times 5x^2$ .

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### Inequalities

- 31 Solve  $3x - 4 > 11$ .
- 32 Solve  $-2x + 6 \leq 20$ .
- 33 Solve  $5x + 1 \geq 3x + 9$ .
- 34 Solve  $4(x - 2) < 2x + 6$ .
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### Substitution into formulas

- 35 The formula for simple interest is  $I = Prt$ . Find  $I$  when  $P = 2,000$ ,  $r = 0.04$ ,  $t = 3$ .
- 36 The perimeter of a rectangle is  $P = 2(l + w)$ . Find  $P$  when  $l = 15$  and  $w = 9$ .
- 37 If  $v = u + at$ , find  $v$  when  $u = 5$ ,  $a = 3$ ,  $t = 4$ .
- 38 The area of a triangle is  $A = \frac{1}{2}bh$ . Find  $A$  when  $b = 14$  and  $h = 9$ .
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### Translating words into equations

- 39 Twice a number, plus 7, equals 25. Find the number.
- 40 A number decreased by 8 is equal to three times the number, decreased by 20. Find the number.
- 41 The sum of three consecutive integers is 84. Find the integers.
- 42 Five more than three times a number is the same as 10 less than five times the number. Find the number.
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### Word problems

- 43 A rectangle has length  $(x + 3)$  cm and width  $x$  cm. This question has two parts.
- a If the area is  $40 \text{ cm}^2$ , write an equation in  $x$ .
- b Solve for  $x$ .
- 44 A father is 3 times as old as his son. In 12 years, he will be twice as old as his son. Find their current ages.
- 45 Tickets cost 25 riyals for adults and 15 riyals for children. A group of 10 people paid 220 riyals in total. How many adults were in the group?

- 46 The sum of two numbers is 45, and their difference is 11. Find the two numbers.
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### MCQ — Evaluating expressions

- 47 If  $x = 6$ , evaluate  $4x - 9$ .
- a. 33
  - b. 24
  - c.  $-15$
  - d. 15
- 48 If  $a = -2$  and  $b = 3$ , evaluate  $a^2 + 2ab$ .
- a.  $-16$
  - b. 8
  - c.  $-4$
  - d.  $-8$
- 49 If  $x = 5$  and  $y = -1$ , evaluate  $(x + y)^2 - xy$ .
- a. 16
  - b. 11
  - c. 21
  - d.  $-5$

50 If  $p = 3$ , evaluate  $2p^3 - 5p$ .

a. 49

b. 39

c. 33

d. 69

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### MCQ — Solving linear equations

51 Solve  $\frac{x}{3} + 4 = 9$ .

a.  $x = 27$

b.  $x = 15$

c.  $x = 5$

d.  $x = 39$

52 Solve  $4(x + 2) = 3x + 15$ .

a.  $x = 1$

b.  $x = 23$

c.  $x = -7$

d.  $x = 7$

53 Solve  $0.5x - 3 = 7$ .

a.  $x = 20$

b.  $x = 8$

c.  $x = -20$

d.  $x = 14$

54 Solve  $\frac{2x-1}{3} = 5$ .

a.  $x = 4$

b.  $x = 7$

c.  $x = 16$

d.  $x = 8$

55 Solve  $6 - 2x = 3x - 9$ .

a.  $x = -3$

b.  $x = 3$

c.  $x = 15$

d.  $x = 1$

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### MCQ — Expanding and simplifying

56 Expand  $(x + 7)^2$ .

a.  $x^2 + 7x + 49$

b.  $x^2 + 14x + 14$

c.  $x^2 + 14x + 49$

d.  $x^2 + 49$

57 Expand  $(x - 5)^2$ .

a.  $x^2 + 10x + 25$

b.  $x^2 - 5x + 25$

c.  $x^2 - 10x + 25$

d.  $x^2 - 25$

58 Expand  $(3x - 2)(3x + 2)$ .

a.  $9x^2 + 4$

b.  $9x^2 - 12x - 4$

c.  $9x^2 - 4$

d.  $6x^2 - 4$

59 Simplify  $2(x + 3) - 3(x - 1)$ .

a.  $5x + 9$

b.  $-x + 9$

c.  $x + 9$

d.  $-x + 3$

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### MCQ — Factoring

60 Factor  $3x^2 + 12x$ .

a.  $3(x^2 + 4x)$

b.  $x(3x + 12)$

c.  $3x(x + 12)$

d.  $3x(x + 4)$

61 Factor  $x^2 - 11x + 30$ .

a.  $(x - 5)(x - 6)$

b.  $(x - 2)(x - 15)$

c.  $(x + 5)(x + 6)$

d.  $(x - 3)(x - 10)$

62 Factor  $x^2 + 3x - 18$ .

a.  $(x + 2)(x - 9)$

b.  $(x - 6)(x + 3)$

c.  $(x + 9)(x - 2)$

d.  $(x + 6)(x - 3)$

63 Factor  $4x^2 - 25$ .

a.  $(2x - 25)(2x + 1)$

b.  $(2x - 5)(2x + 5)$

c.  $(2x - 5)^2$

d.  $(4x - 5)(x + 5)$

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### MCQ — Solving quadratic equations

64 Solve  $x^2 - 9x + 18 = 0$ .

a.  $x = -3$  or  $x = -6$

b.  $x = 1$  or  $x = 18$

c.  $x = 2$  or  $x = 9$

d.  $x = 3$  or  $x = 6$

65 Solve  $x^2 + 2x - 24 = 0$ .

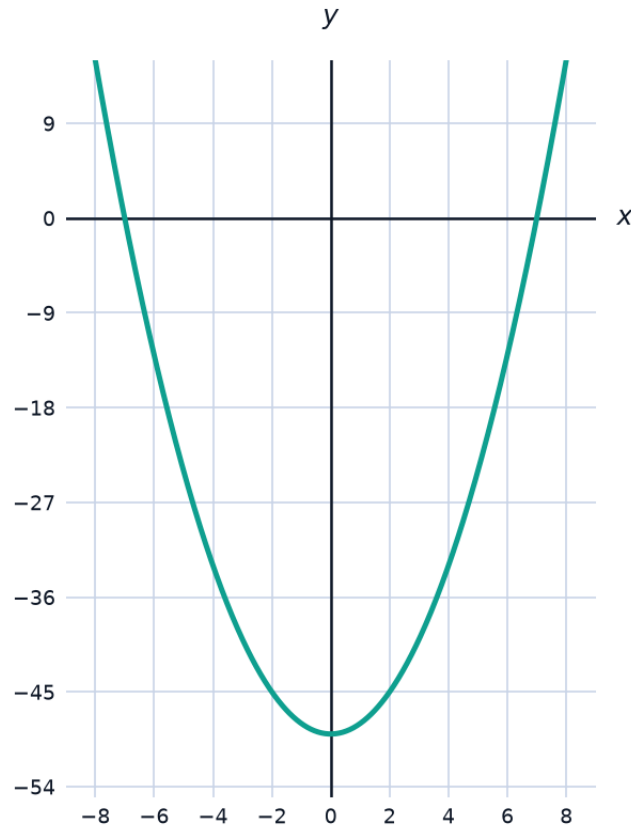
a.  $x = 6$  or  $x = -4$

b.  $x = -2$  or  $x = 12$

c.  $x = -6$  or  $x = 4$

d.  $x = -3$  or  $x = 8$

- 66 The graph shows  $y = x^2 - 49$ . What are its roots (where the curve crosses the  $x$ -axis)?



- a.  $x = 7$  only
  - b.  $x = 49$
  - c.  $x = 7$  or  $x = -7$
  - d.  $x = -7$  only
- 67 Solve  $2x^2 - 8x = 0$ .
- a.  $x = 8$
  - b.  $x = 2$  or  $x = 4$
  - c.  $x = 0$  or  $x = 4$
  - d.  $x = 0$  or  $x = -4$

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## MCQ — Systems of two equations

**68** Solve the system:  $x + 2y = 8$ ,  $x - y = 2$ .

**a.**  $x = 2$ ,  $y = 3$

**b.**  $x = 6$ ,  $y = 1$

**c.**  $x = 4$ ,  $y = 2$

**d.**  $x = 8$ ,  $y = 0$

**69** Solve the system:  $3x - y = 7$ ,  $x + y = 5$ .

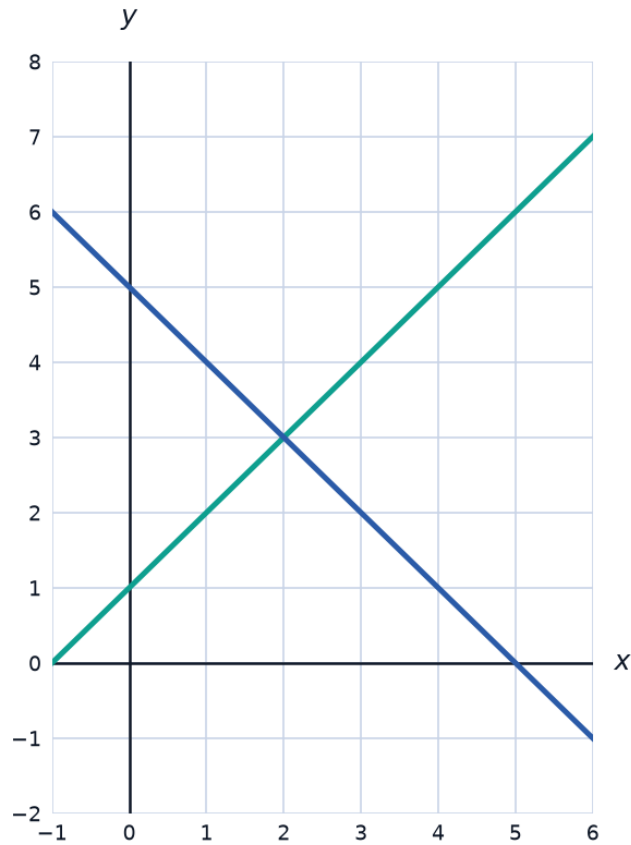
**a.**  $x = 2$ ,  $y = 3$

**b.**  $x = 1$ ,  $y = 4$

**c.**  $x = 3$ ,  $y = 2$

**d.**  $x = 4$ ,  $y = 1$

70 The graph shows two lines. What is their point of intersection?



a. (1, 4)

b. (0, 5)

c. (2, 3)

d. (3, 2)

71 Solve the system:  $y = 2x - 1$ ,  $y = x + 3$ .

a.  $x = -4$ ,  $y = -5$

b.  $x = 2$ ,  $y = 3$

c.  $x = 7$ ,  $y = 4$

d.  $x = 4$ ,  $y = 7$

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## MCQ — Laws of exponents

72 Simplify  $b^7 \div b^2$ .

a.  $b^{3.5}$

b.  $b^{14}$

c.  $b^9$

d.  $b^5$

73 Simplify  $(2x^3)^2$ .

a.  $4x^6$

b.  $2x^9$

c.  $4x^5$

d.  $2x^6$

74 Simplify  $x^0 \cdot x^4$  ( $x \neq 0$ ).

a. 1

b.  $x^5$

c.  $x^0$

d.  $x^4$

75 Simplify  $m^{-2}$ .

a.  $-\frac{1}{m^2}$

b.  $\frac{1}{m^2}$

c.  $m^2$

d.  $-m^2$

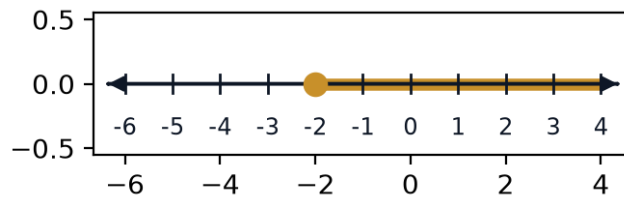
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## MCQ — Inequalities

- 76 The number line shows the solution set of an inequality. Which inequality matches?



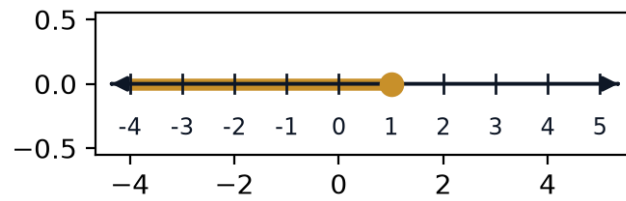
- a.  $x \geq 4$
  - b.  $x \leq 4$
  - c.  $x > 4$
  - d.  $x < 4$
- 77 The number line shows the solution set of an inequality. Which inequality matches?



- a.  $x \leq -2$
  - b.  $x < -2$
  - c.  $x \geq -2$
  - d.  $x > -2$
- 78 Solve  $-3x + 7 \leq 22$ .

- a.  $x \geq -5$
- b.  $x \leq -5$
- c.  $x \leq 5$
- d.  $x \geq 5$

79 The number line shows the solution set of an inequality. Which inequality matches?

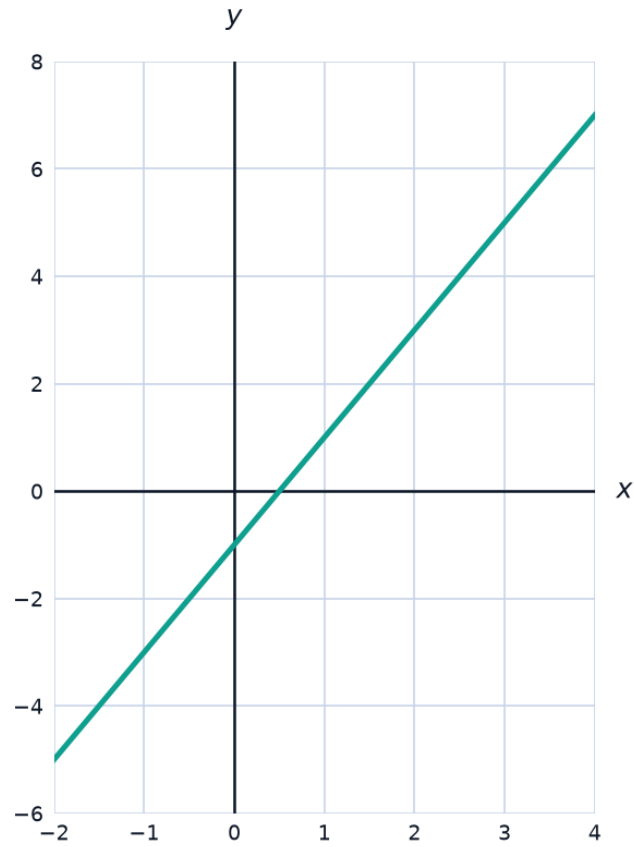


- a.  $x > 1$
- b.  $x \geq 1$
- c.  $x \leq 1$
- d.  $x < 1$

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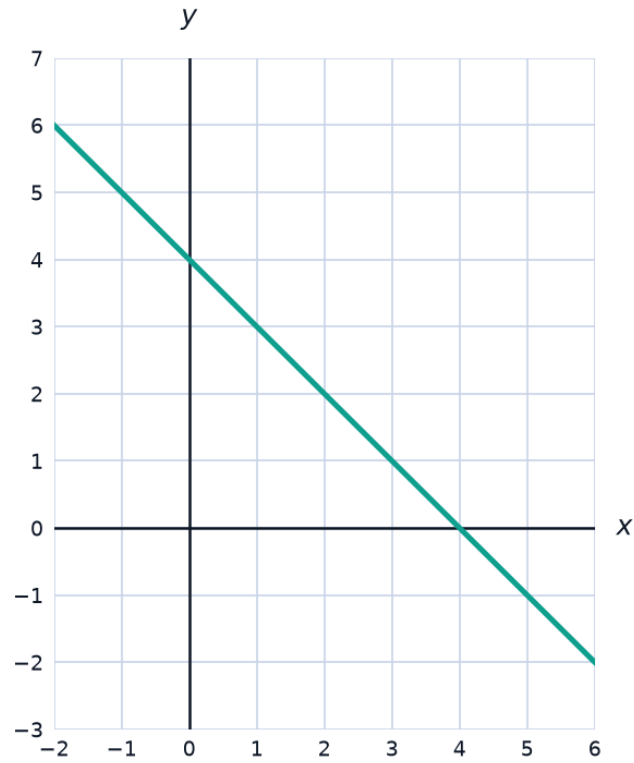
### MCQ — Reading function graphs

80 The graph shows a line. What is its slope?



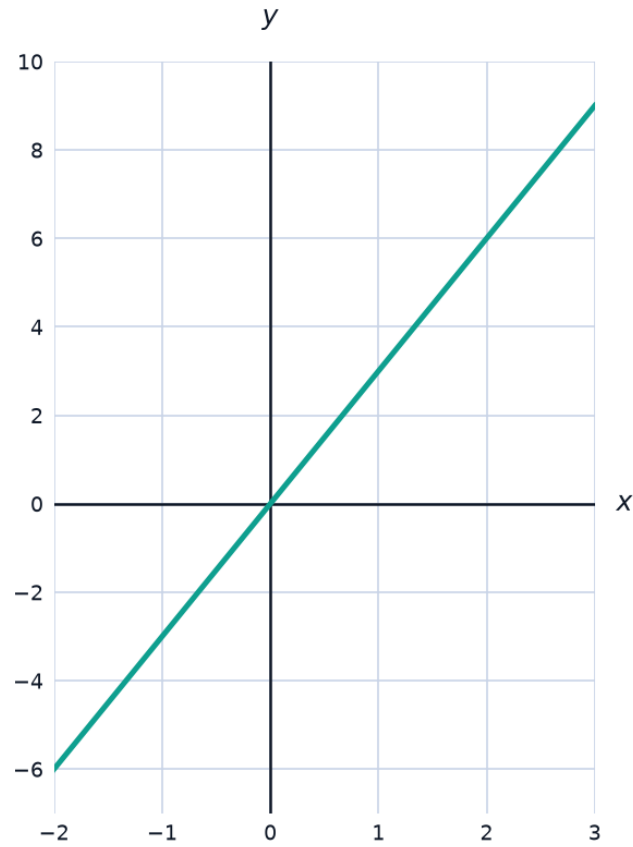
- a.  $\frac{1}{2}$
- b. 2
- c. -2
- d. -1

81 What is the y-intercept of the line shown?



- a. 4
- b. 0
- c. -4
- d. 1

82 What is the equation of the line shown?



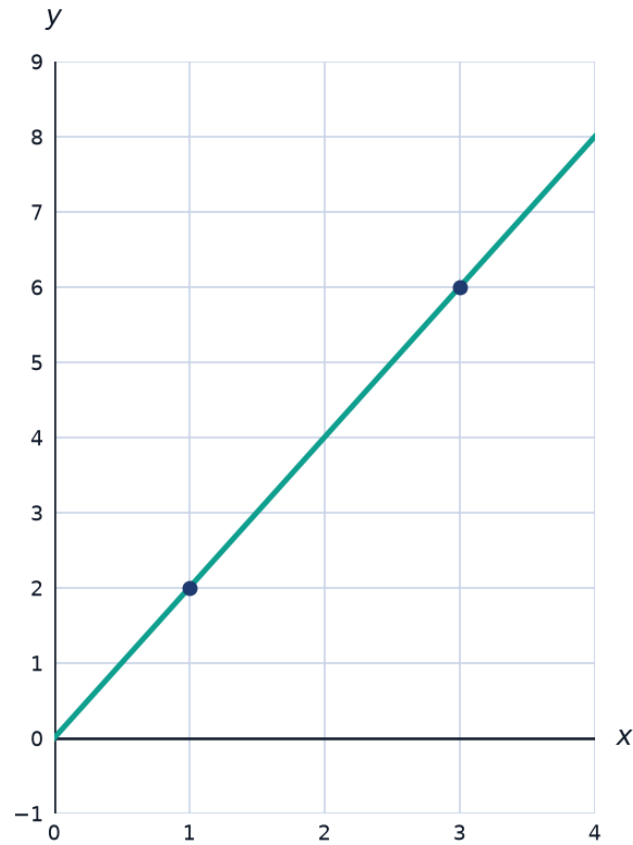
a.  $y = 3x + 1$

b.  $y = x + 3$

c.  $y = \frac{1}{3}x$

d.  $y = 3x$

- 83 Find the slope of the line through the two marked points.



- a. 2
- b.  $\frac{1}{2}$
- c. -2
- d. 4

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### MCQ — Substitution into formulas

- 84 Given  $A = \frac{1}{2}bh$ , find  $A$  when  $b = 10$  and  $h = 6$ .

- a. 16
- b. 8
- c. 60
- d. 30

85 Given  $KE = \frac{1}{2}mv^2$ , find  $KE$  when  $m = 4$  and  $v = 3$ .

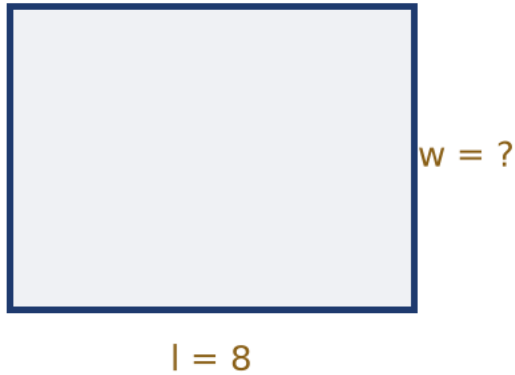
a. 18

b. 12

c. 24

d. 36

86 The rectangle shown has area  $A = lw = 48$  and length  $l = 8$ . Find the width  $w$ .



a. 8

b. 6

c. 384

d. 40

87 Given  $F = ma$ , find  $F$  when  $m = 7$  and  $a = 5$ .

a. 35

b. 75

c. 12

d. 2.4

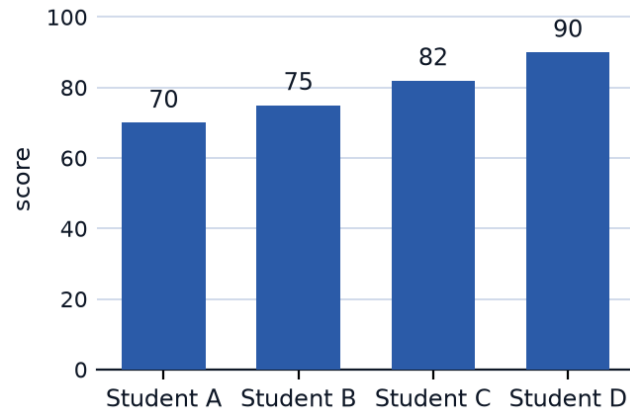
## MCQ — Translating words into equations

- 88** Three times a number, minus 5, equals twice the number plus 7. Find the number.
- a. 19
  - b. 2
  - c.  $-12$
  - d. 12
- 89** The sum of two consecutive even integers is 54. Find the larger integer.
- a. 28
  - b. 27
  - c. 29
  - d. 26
- 90** Five less than three times a number is 25. Find the number.
- a. 10
  - b. 20
  - c. 5
  - d. 15
- 91** The perimeter of a rectangle is 54 cm. The length is 3 cm more than the width. Find the width.
- a. 24
  - b. 12
  - c. 15
  - d. 9

## MCQ — Word problems

- 92 A number increased by 20% equals 72. Find the number.
- a. 90
  - b. 52
  - c. 60
  - d. 86.4
- 93 Ahmed is twice as old as his sister. In 5 years, the sum of their ages will be 40. Find Ahmed's current age.
- a. 20
  - b. 10
  - c. 15
  - d. 25
- 94 A positive number plus its square is 30. Find the number.
- a. 30
  - b.  $-6$
  - c. 6
  - d. 5

- 95 The bar chart shows four students' test scores. Whose score is exactly 12 more than 70?



- a. Student B
  - b. Student D
  - c. Student C
  - d. Student A
- 96 The product of two consecutive integers is 132. Find the larger integer.
- a. 13
  - b. 12
  - c. 10
  - d. 11